GIDEON ATOYAH

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PROFILE

Biomedical Engineering graduate (First Class Honours, KNUST) with hospital clinical engineering experience and independent research in sensor-based medical devices, embedded systems, health software, and deep learning for medical imaging. Currently developing a synthetic-data pipeline and segmentation model for orthopaedic implant localization in X-ray imaging. Seeking PhD and research-based graduate study in medical imaging, machine learning for healthcare, bioinstrumentation, and biomechanics.

EDUCATION

BSc Biomedical Engineering — First Class Honours *2021 – 2025*
Kwame Nkrumah University of Science and Technology (KNUST), Kumasi, Ghana

- Cumulative weighted average (CWA): **74.35/100** over 152 credit hours; degree classification First Class (70–100%). Semester weighted averages of 80.48 (Year 2, Semester 1) and 80.81 (Year 4, Semester 2); cumulative average rose from 67.63 (Year 1) to 74.35.
- **Highest marks:** C Programming (95), Object-Oriented Programming (93), Semiconductor Devices (91), Bioinstrumentation I (88), Digital Signal Processing (88), Biomedical Engineering Design I & II (88, 88), Microprocessors (84), Project I & II (83, 84), Medical Device Regulation (80).
- **Relevant coursework:** Medical Imaging I & II, Bioinstrumentation I & II, Biosignal Processes and Analysis, Biomedical Optics, Biomechanics I & II, Digital Signal Processing, Signals and Systems, Embedded Systems, Microprocessors, Numerical Methods, Probability and Statistics, Biomaterials I & II, Medical Device Regulation, Research Methods, Biomedical Engineering Design I & II.

W.A.S.S.C.E (General Science) *2018 – 2021*
St Hubert Seminary Senior High School, Kumasi, Ghana

RESEARCH AND PROJECT EXPERIENCE

Synthetic X-ray Generation & Deep Learning Segmentation for THR Assessment *Jan 2026 – Present*
Independent Research

- Built a synthetic X-ray pipeline: inserted 3D implant models into TCIA CT volumes (3D Slicer) and rendered synthetic X-rays via DeepDRR, with a custom tool to randomize noise, sharpness, and blur.
- Training a U-Net segmentation model on the synthetic data and testing generalization on OrthoNet (real clinical X-rays); next phase adds HRNet for landmark detection and clinical parameter computation.
- Status: ongoing, model training in progress.
- **Code:** Kaggle notebook (DeepDRR pipeline).

FSR-Based In-Cast Pressure Monitoring System *2024 – 2025*
Final-Year Thesis (BME 497/498) | Supervisor: Dr Isaac Acquah | Group Leader

- Designed a system to objectively measure cast-induced skin pressure using 8 FSR sensors, an Arduino Nano signal chain, Bluetooth transfer, and a FastAPI backend.
- Built a PyQt5 desktop app visualizing pressure as a real-time 3D heat-map. Defended before a panel; BME 497/498 graded 83% and 84%.
- **Thesis paper:** Full-text PDF (Google Drive); also presented as a poster at KNUST Technology Week (Aug 2025).

National Cleft Registry System *2026*
Independent Software Project — solo-built

- Built a two-hub (Next.js/TypeScript/SQLite) system for tracking cleft lip/palate patients across Ghanaian hospitals, with offline-first sync to a central hub at Komfo Anokye Teaching Hospital.
- Independently fixed critical bugs (data loss, broken sync, unauthenticated endpoints) and packaged the system for Windows hospital deployment.
- **Source code:** GitHub repository (national-cleft-register).

Smart Drug Delivery Dispenser (Arduino Prototype) *2023*

Group Course Project, COE 381 Microprocessors (84%) | Lead Programmer

- Designed and built a prototype medication dispenser from an Arduino microcontroller, LCD, ultrasonic distance sensor, push buttons, and a cardboard housing, to support timed, supervised drug intake.
- Operation: the user sets the dosing time with the buttons and LCD; at that time an alarm sounds continuously until the user moves within range of the ultrasonic sensor, which releases the dose and resumes the timer for the next cycle.
- Solely responsible for programming the entire system (timer logic, user interface, alarm control, and sensor-triggered dispensing).
- Video: Demonstration video (Google Drive).

PRESENTATIONS

Poster Presentation: FSR-Based In-Cast Pressure Monitoring System *Aug 2025*

Technology Week (TWEEK), Department of Computer Engineering, KNUST

- Presented my final-year thesis poster to students and business partners at the department-hosted technology week.

WORK EXPERIENCE

Biomedical engineer | GS Medikals and Birim Energy *Sep 2026 – present*

- Building of an oxygen production room and pipe integration at Holy Family Catholic Hospital, Berekum
- Installation of oxygen production equipment, Bed Head Units and Ceiling pendants at Tamale Teaching Hospital and Komfo Anokye Teaching Hospital.

National Service Personnel | Komfo Anokye Teaching Hospital *Nov 2025 – Aug 2026*

- Supported hands-on installation of patient monitors, C-arms, dialysis machines, ventilators, and anaesthesia machines, alongside manufacturer engineers (Iberica, Nivaansh) on specialized equipment.
- Performed theatre preparation, device calibration, and preventive maintenance; generated standardized PM protocol documentation (ReportLab) and delivered an internal presentation on ventilators and anaesthesia machines.

Intern | Komfo Anokye Teaching Hospital *Sept–Dec 2022 & Sept–Dec 2023*

- Diagnosed and resolved faults in BP apparatus, suction machines, glucometers, and infusion pumps; developed proficiency fixing power-related issues in AC-powered medical devices.

LEADERSHIP

- Group Leader: FSR-Based In-Cast Pressure Monitoring System (final-year thesis, BME 497/498).
- Secretary: Presbyterian Church Choir.

CERTIFICATIONS AND ONLINE COURSEWORK

- **Coursera (imaging and machine learning):** Introduction to Machine Learning; Fundamentals of Biomedical Imaging 1; Medical Image Processing.
- **Coursera (biomechanics and robotics):** Musculoskeletal Anatomy; Anthropometry, Biomechanics, and Motor Skills in User Design; Theoretical Bases of Health and Injury Management; Modern Robotics.
- Introduction to AI : ALX (2024); AI and Machine Learning: Dr Isaac Acquah (2024); Dialysis Training: Dr Isaac Acquah (2024).
- Python Programming: Udemy (2024–2025); Frontend Development: CodTech IT Solutions (2024).
- SOLIDWORKS Essential Training & FEA Simulation, Project Management (Healthcare Projects, Asana): LinkedIn Learning / PMI (2026).

SKILLS

- Deep learning & imaging: Python, PyTorch, DeepDRR, 3D Slicer: Synthetic data generation and segmentation model training.
- Programming & software: Python, TypeScript, C, FastAPI, SQLite, InfluxDB, Next.js: full-stack development and deployment.
- Instrumentation & CAD: sensor/circuit design (EasyEDA, Proteus), embedded systems (Arduino), SolidWorks 3D modelling and FEA.
- Clinical equipment: maintenance, calibration, and installation across patient monitors, ventilators, anaesthesia and dialysis machines, C-arms.